

NON-MF GROUPS

SAUERS

We show that not every countable group is MF, where MF means admitting finite-dimensional unitary models that are asymptotically multiplicative and asymptotically separate nonidentity elements in operator norm. The main tool is a one-sided compression criterion that places normal Kazhdan subgroups in the MF radical. Applied to the binary Leavitt group, it shows that the finitely generated simple property-(T) group $H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$ has trivial image under every homomorphism to an MF group; consequently its reduced group C^* -algebra is a separable stably finite C^* -algebra that is not MF. A second application gives a finitely presented torsion-free group with property (T) that has trivial image under every homomorphism to an MF group, and every nontrivial quotient of it has the same property.

**DRAFT — NOT READY FOR PUBLIC
CONSUMPTION**

This manuscript is actively being revised and has not been released for public consumption.

1. INTRODUCTION

Not every countable group is MF. The counterexample is

$$H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2)).$$

The group H is finitely generated, nontrivial, simple, and has property (T), but every homomorphism from H to an MF group is trivial. In particular, H is not MF. Its reduced group C^* -algebra $C_r^*(H)$ is separable and stably finite, but is not MF. The group is not new: by the isomorphisms of Khanh–Thanh, H is the unit group of the binary Leavitt algebra [K–T], with which OpenAI constructed the first nonsofic group [OAI].

More precisely, a countable group G is *MF* if there are positive integers d_n and maps $V_n: G \rightarrow \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \quad (g \in G \setminus \{1\}).$$

2020 *Mathematics Subject Classification*. Primary 46L05; Secondary 20C07, 16S50, 20G35.

Key words and phrases. MF groups, norm matrix coronas, MF radicals, property (T), asymptotic representations, Leavitt algebra.

This is the operator norm notion of MF group introduced by Carrión–Dadarlat–Eckhardt [CDE]. Here, MF group always means this operator-norm local-approximation property. The strong-convergence convention also requires the matrix models to recover the norms of the left regular representation [GKM, Sch].

Equivalently, G embeds in the unitary group of the norm matrix corona associated to the sequence $\mathbf{d} = (d_n)$,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here

$$\bigoplus_n M_{d_n}(\mathbb{C}) = \{(x_n) \in \prod_n M_{d_n}(\mathbb{C}) : \|x_n\| \rightarrow 0\}$$

is the c_0 -direct sum.

A separable C^* -algebra is called MF if it embeds as a C^* -subalgebra of a norm matrix corona. For every countable group G , the reduced group algebra $C_r^*(G)$ is separable and has a faithful canonical trace, hence is stably finite. A non-MF group therefore gives a separable stably finite reduced group C^* -algebra that is not MF (see the proof of Theorem B).

For a countable group G , define its MF radical by

$$\text{Rad}_{\text{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

We call a homomorphism $G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ a *corona homomorphism*. Proposition 2.1 shows that G is MF precisely when $\text{Rad}_{\text{MF}}(G)$ is trivial. If $\text{Rad}_{\text{MF}}(G) = G$, every homomorphism from G to an MF group is trivial.

For a subgroup $L \leq G$, put

$$\text{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}$$

and define its *compression-centralizer defect* by

$$(1) \quad \mathfrak{D}_G(L) = \langle\langle [ucu^{-1}, \ell] : u \in \text{Comp}_G(L), c \in C_G(L), \ell \in L \rangle\rangle_G.$$

Theorem A (one-sided compression criterion). *Let G be countable and let $L \leq G$ have property (T). If $K \trianglelefteq G$ has property (T) and $K \leq \mathfrak{D}_G(L)$, then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

In particular, a nontrivial such K makes G non-MF. If G has property (T) and $\mathfrak{D}_G(L) = G$, then $\text{Rad}_{\text{MF}}(G) = G$.

We sketch the argument. Corollary 2.6 gives $\mathfrak{D}_G(L) \leq R_{\infty \rightarrow 2}(G)$. That is, every operator norm asymptotic representation of G is asymptotically trivial on $\mathfrak{D}_G(L)$ in normalized Hilbert–Schmidt norm. This holds because conjugation by the compressor moves the Kazhdan projection of L below itself, and stable finiteness of the corona forces equality (Section 2.3). If a corona homomorphism were nontrivial on K , the Kazhdan projection of K would cut out a G -invariant corner on which K has no nonzero fixed vectors (Lemma 2.7); by the Kazhdan inequality some $s_0 \in K$ would then

stay bounded away from the identity in normalized Hilbert–Schmidt norm along a subsequence, contradicting $s_0 \in \mathfrak{D}_G(L)$.

Theorem 2.2 gives a separate finite-dimensional vanishing result: every finite-dimensional linear representation of G over any field kills $\mathfrak{D}_G(L)$.

Property (T) is not itself an obstruction to MF approximation: every residually finite group is MF, including many Kazhdan groups. In the argument above, property (T) controls the fixed vectors of an exact corona representation. The one-sided compression forces the corresponding Kazhdan projection to commute with the compressor, which makes the Hilbert–Schmidt commutant invariant.

We next apply Theorem A to the group H .

Let $L_{\mathbb{F}_2}(1, 2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$ -algebra generated by s_0, s_1, t_0, t_1 subject to

$$(2) \quad t_i s_j = \delta_{ij}, \quad s_0 t_0 + s_1 t_1 = 1.$$

For a unital ring R , write $e_{ij}(a) = 1 + aE_{ij}$ and let $\text{EL}_n(R)$ be the subgroup of $\text{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem B (the binary Leavitt group). *Put $R = L_{\mathbb{F}_2}(1, 2)$ and $H = \text{EL}_{12}(R)$. Then H is finitely generated, nontrivial, simple, has property (T), and*

$$\text{Rad}_{\text{MF}}(H) = H.$$

Equivalently, every homomorphism from H to an MF group is trivial. In particular, H is non-MF. Its reduced group C^ -algebra $C_r^*(H)$ is separable and stably finite, but is not MF.*

Inside H , an explicit element τ satisfies $\tau L \tau^{-1} \leq L$ for the subgroup $L \cong \text{EL}_3(R)$ in the upper left. The element $c = e_{34}(1)$ centralizes L , and for $\ell = e_{12}(1)$ the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

The relation $t_1(s_1 t_1)s_1 = 1$ and the Steinberg commutator relations show directly that d normally generates H . Since $d \in \mathfrak{D}_H(L)$ and $\mathfrak{D}_H(L)$ is normal, it follows that $\mathfrak{D}_H(L) = H$. Therefore Theorem A gives $\text{Rad}_{\text{MF}}(H) = H$.

Theorem C (a torsion-free group with full MF radical). *There is a two-generated, finitely presented, torsion-free, acylindrically hyperbolic group Q with property (T) and*

$$\text{Rad}_{\text{MF}}(Q) = Q.$$

Every nontrivial quotient of Q also equals its own MF radical; in particular, no nontrivial quotient of Q is MF.

The group Q is a small-cancellation quotient, in the sense of Hull [Hu], of the torsion-free property-(T) group of Fournier-Facio [FF, §2], chosen so that the compression defect normally generates the quotient (Section 4). Thus Theorem A also applies to Q .

Relation to prior work. Blackadar and Kirchberg developed the notions of MF and NF algebras and proved that a separable C^* -algebra is NF if and only if it is nuclear and MF [BK]. It also asked whether every separable nuclear stably finite C^* -algebra is quasidiagonal. This was a question about nuclear C^* -algebras, not about arbitrary groups.

The group property used here was introduced by Carrión–Dadarlat–Eckhardt [CDE]. If every separable stably finite C^* -algebra were MF, then every countable group would be MF because its reduced group C^* -algebra is separable and stably finite. Later papers on approximation of groups relate the resulting operator norm problem to Kirchberg’s question [LO, Th]. The negative solution of the Connes embedding problem [JNVWY] yields separable stably finite C^* -algebras that are not MF, as Goldbring and Hart observe [GH, Proposition 6.1 and Remark 6.2]. Theorem B gives such an example among reduced group C^* -algebras. Korchagin subsequently developed permanence and local approximation results for MF groups [Kor]. Shulman proves that $A *_C A$ is MF whenever A is a separable MF C^* -algebra and $C \subseteq A$ is a C^* -subalgebra [Sh, Theorem 10]. She also proves that the full group C^* -algebras of amalgamated free products of amenable groups and of certain semidirect products are MF [Sh]. Nonapproximability was known for the unnormalized Schatten p -norms with $1 < p < \infty$ [DGLT, LO]. Bachner–Dogon–Lubotzky later proved nonapproximability for $p = 1$ [BDL]. The operator norm case $p = \infty$ remained open. They also identify stability from operator norm to normalized Hilbert–Schmidt norm as a possible route to a non-MF group [BDL, Proposition 1.5]; that is the route taken here.

Leavitt introduced the algebras $L_K(1, n)$ as examples of rings with nonunique module rank [Lev]. The compression used here comes from their defining relations. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser classified the subgroups of GL_n over an exchange ring that are normalized by EL_n [Pre]. His sandwich theorem, together with the simplicity of the binary Leavitt algebra and the equality $Z(R) = \mathbb{F}_2$, implies that $EL_{12}(R)$ is simple (Proposition 3.3). Ershov–Jaikin–Zapirain give property (T) for $EL_n(R)$ whenever $n \geq 3$ and R is a finitely generated unital associative ring [EJZ, Theorem 1.1]. OpenAI used the binary Leavitt unit group to construct the first known nonsofic group [OAI, Chapter 3]. Khanh–Thanh’s matrix-ring isomorphisms and their equality $GL_n(R) = EL_n(R)$ identify that group with the group H studied here [K–T, Proposition 4.2 and Corollary 4.4]. We apply the Leavitt compression to operator norm approximation. Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT, Dad].

2. ONE-SIDED COMPRESSION AND THE MF RADICAL

2.1. The MF radical. The image of a corona homomorphism from G is a countable subgroup of a corona unitary group, hence MF; conversely, every

countable MF group embeds in the unitary group of a norm matrix corona. Thus $\text{Rad}_{\text{MF}}(G)$ is also the intersection of the kernels of all homomorphisms from G to MF groups.

Proposition 2.1 (basic properties of the MF radical). *For every countable group G , the subgroup $\text{Rad}_{\text{MF}}(G)$ is fully invariant, the quotient $G/\text{Rad}_{\text{MF}}(G)$ is MF, and G is MF if and only if $\text{Rad}_{\text{MF}}(G)$ is trivial.*

Proof. An intersection of kernels is normal. If α is an endomorphism of G , $x \in \text{Rad}_{\text{MF}}(G)$, and π is a corona homomorphism from G , then $\pi \circ \alpha$ kills x . Hence $\pi(\alpha(x)) = 1$ for every π , which proves full invariance.

Put $R = \text{Rad}_{\text{MF}}(G)$. The trivial group is MF. If G/R is nontrivial, let $x \in G/R$ be nontrivial and choose $g \in G$ mapping to x . Then $g \notin R$. Some homomorphism from G to an MF group therefore separates g from the identity. It kills R and descends to a homomorphism from G/R that separates x . Thus G/R is residually MF, and hence MF by Korchagin [Kor, Proposition 6].

If G is MF, a corona embedding of G has trivial kernel, so $\text{Rad}_{\text{MF}}(G) = 1$. Conversely, if $\text{Rad}_{\text{MF}}(G) = 1$, then $G = G/\text{Rad}_{\text{MF}}(G)$ is MF. $\square$

2.2. Finite dimension.

Theorem 2.2 (finite-dimensional commutant rigidity). *Let G be a group, let $L \leq G$, and suppose $u \in G$ satisfies $uLu^{-1} \leq L$. If k is a field, V is a finite-dimensional k -vector space, and $\rho: G \rightarrow \text{GL}(V)$ is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

where $\rho(L)'$ is the commutant of $\rho(L)$ in $\text{End}_k(V)$. Consequently, if $c \in C_G(L)$, then

$$\rho([ucu^{-1}, \ell]) = 1 \quad (\ell \in L).$$

Proof. Put $C = \rho(L)'$. If $x \in C$, $h \in L$, and $h' = uhu^{-1} \in L$, then

$$\begin{aligned} \rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u) \\ &= \rho(u)^{-1}x\rho(h')\rho(u) \\ &= \rho(u)^{-1}x\rho(u)\rho(h). \end{aligned}$$

Thus $\rho(u)^{-1}C\rho(u) \subseteq C$. Conjugation by $\rho(u)^{-1}$ is injective, and C is finite-dimensional. Its restriction to C is therefore surjective, so the inclusion is equality. Conjugating by $\rho(u)$ gives $\rho(u)C\rho(u)^{-1} = C$.

If $c \in C_G(L)$, then $\rho(c) \in C$, hence $\rho(ucu^{-1}) \in C$. Therefore

$$\rho([ucu^{-1}, \ell]) = 1 \quad (\ell \in L).$$

$\square$

Finite dimensionality is essential: an injective linear self-map of an infinite-dimensional commutant need not be surjective.

2.3. Kazhdan transport in normalized Hilbert–Schmidt norm. For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \text{tr}_d(a^*a)^{1/2}, \quad \text{tr}_d = \frac{1}{d} \text{Tr}.$$

An *operator norm asymptotic representation* of a countable group G is a sequence of maps $V_n: G \rightarrow \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \quad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \longrightarrow 0\}.$$

The operator norm defects also tend to zero in normalized Hilbert–Schmidt norm, so $K_2(V)$ is a normal subgroup of G . Define

$$(3) \quad R_{\infty \rightarrow 2}(G) = \bigcap_V K_2(V),$$

where V ranges over all operator norm asymptotic representations of G .

For $L \leq G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \left| \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|[V_n(\ell), x_n]\|_2 \longrightarrow 0 \quad (\ell \in L) \end{array} \right. \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\text{Ad}(V(g))$, of the bounded matrix sequences.

Lemma 2.3. *For every sequence of positive integers (m_n) , the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Suppose that $v^*v = 1$ in the corona, and let (x_n) be a bounded lift of v . Then $x_n^*x_n \rightarrow 1$ in norm. For all sufficiently large n , the matrix x_n is invertible, and the unitary $x_n(x_n^*x_n)^{-1/2}$ differs from x_n by $o(1)$. Thus v is represented by unitaries and $vv^* = 1$. The same argument applies to every matrix amplification of the corona. If A is a stably finite unital C^* -algebra, equivalent projections $p \leq q$ satisfy $p = q$. Indeed, if $w^*w = q$ and $ww^* = p$, then $w \in qAq$ is an isometry in the corner. The corner is finite because $w + (1 - q)$ is an isometry in A . Consequently $ww^* = q$, and hence $p = q$. $\square$

Lemma 2.4 (one-sided order for the Kazhdan projection). *Let L have property (T), let B be a unital C^* -algebra, and let $\pi: L \rightarrow \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C_{\max}^*(L) \rightarrow B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \leq P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. The represented projection P is the orthogonal projection onto $\text{Fix } \pi(L)$. If ξ is L -fixed and $\ell \in L$, then

$$\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi,$$

because $U\pi(\ell)U^*$ belongs to $\pi(L)$. Hence $U^*\text{Fix } \pi(L) \subseteq \text{Fix } \pi(L)$, so the range projection U^*PU is dominated by P in $B(\mathcal{H})$. Faithfulness of the representation gives the same projection inequality in B . $\square$

Theorem 2.5 (one-sided Kazhdan transport). *Let $L \leq G$ have property (T), and let $u \in G$ satisfy $uLu^{-1} \leq L$. Let (V_n) be an operator norm asymptotic representation of G . Then*

$$\text{Ad}(V(u))(\mathcal{C}_2(V, L)) = \mathcal{C}_2(V, L).$$

Proof. Fix $x = (x_n) \in \mathcal{C}_2(V, L)$ and let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the ultraproduct of Hilbert spaces $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation σ of G on $\mathcal{K}_\omega$. Indeed, for unitaries A, B one has

$$\|\text{Ad}(A) - \text{Ad}(B)\| \leq 2\|A - B\|,$$

so the conjugation actions are multiplicative modulo c_0 because the multiplicative defects of (V_n) tend to zero in operator norm. The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\tilde{\sigma}(g)$ be the class of the coordinate operators $\text{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C})) / \bigoplus_n B(M_{d_n}(\mathbb{C})).$$

After choosing matrix units on each $M_{d_n}(\mathbb{C})$, this is a norm matrix corona with coordinate sizes d_n^2 . By the preceding estimate, $g \mapsto \tilde{\sigma}(g)$ is a homomorphism $G \rightarrow \mathcal{U}(\mathcal{B})$. Its restriction to L therefore extends to a $*$ -homomorphism $C_{\max}^*(L) \rightarrow \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection [AW]. Under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\text{Fix } \sigma(L)$. The projection identity below is proved inside $\mathcal{B}$; the representation of $\mathcal{B}$ on $\mathcal{K}_\omega$ is used only afterwards. Since $uLu^{-1} \leq L$, the unitary $U = \tilde{\sigma}(u)$ satisfies the hypothesis of Lemma 2.4, and hence $U^*PU \leq P$ inside $\mathcal{B}$. The two projections are unitarily equivalent. Lemma 2.3 therefore gives $U^*PU = P$, so U commutes with P . Hence both $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. Since ω was arbitrary, the corresponding Hilbert–Schmidt commutators converge to zero along the full sequence. Thus $\text{Ad}(V(u))$ and its inverse both preserve $\mathcal{C}_2(V, L)$, which proves the equality. $\square$

Corollary 2.6. *Let $L \leq G$ have property (T), let $uLu^{-1} \leq L$, and let $c \in C_G(L)$. Then*

$$[ucu^{-1}, \ell] \in R_{\infty \rightarrow 2}(G) \quad (\ell \in L).$$

Proof. Let (V_n) be an operator norm asymptotic representation. If $c \in C_G(L)$, then $(V_n(c)) \in \mathcal{C}_2(V, L)$. By Theorem 2.5, the same is true of $(V_n(u)V_n(c)V_n(u)^*)$. Asymptotic multiplicativity identifies this sequence in operator norm with $(V_n(ucu^{-1}))$; explicitly,

$$\|V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1})\| \longrightarrow 0.$$

Therefore

$$\|V_n([ucu^{-1}, \ell]) - 1\|_2 \longrightarrow 0 \quad (\ell \in L).$$

Intersecting over (V_n) proves the claim. $\square$

2.4. Normal Kazhdan subgroups and the MF radical. Let $K \trianglelefteq G$ have property (T), and let p be the image of its Kazhdan projection under a corona homomorphism of G . Normality of K implies that p commutes with the image of G . Hence $q = 1 - p$ also commutes with that image, and in every nondegenerate representation of the corner $q\mathcal{Q}_{\mathbf{d}}q$ the induced representation of K has no nonzero fixed vectors.

Lemma 2.7 (central corona corners). *Let $\rho: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$, integers $r_n = \text{rank}(q_n)$, unitary identifications $J_n: \mathbb{C}^{r_n} \rightarrow q_n\mathbb{C}^{d_n}$, and an operator norm asymptotic representation*

$$W_n: G \longrightarrow \mathcal{U}(r_n)$$

with $W_n(1) = I_{r_n}$. In the corona over the retained coordinates, the class of $(J_n W_n(g) J_n^)$ is the coordinate restriction of $q\rho(g)$.*

Proof. Lift q to projections q_n , and lift each $\rho(g)$ to unitaries $U_n(g)$, choosing $U_n(1) = I_{d_n}$. The first lift is obtained by spectral rounding; the second is obtained by polar correction, since the two unitarity defects of any lift converge to zero in operator norm. The commutation relation in the corona gives

$$\|q_n U_n(g) - U_n(g) q_n\| \longrightarrow 0 \quad (g \in G).$$

Consequently $q_n U_n(g) q_n$, viewed on $q_n \mathbb{C}^{d_n}$, has both unitarity defects converging to zero. For each fixed g , let $\widetilde{W}_n(g)$ be its polar correction whenever both defects are at most $1/2$, and set $\widetilde{W}_n(g) = q_n$ at the finitely many earlier stages. Then $\widetilde{W}_n(g)$ is unitary in the corner and satisfies

$$\|\widetilde{W}_n(g) - q_n U_n(g) q_n\| \longrightarrow 0.$$

For $g = 1$, the compression is already the identity q_n of the corner, so take $\widetilde{W}_n(1) = q_n$. Since $q \neq 0$, infinitely many q_n are nonzero. Retain and relabel

those coordinates. Put $r_n = \text{rank}(q_n)$, choose a unitary $J_n: \mathbb{C}^{r_n} \rightarrow q_n \mathbb{C}^{d_n}$, and define

$$W_n(g) = J_n^* \widetilde{W}_n(g) J_n \in \mathcal{U}(r_n).$$

Conjugation by J_n preserves operator norms and multiplicative defects. Thus (W_n) is an operator norm asymptotic representation with $W_n(1) = I_{r_n}$. The displayed estimate identifies the class of $(J_n W_n(g) J_n^*)$ with the coordinate restriction of $q\rho(g)$. $\square$

Theorem 2.8 (normal Kazhdan radical theorem). *Let G be countable and let $K \trianglelefteq G$ have property (T) with $K \leq R_{\infty \rightarrow 2}(G)$. Then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

Proof. Suppose that a homomorphism $\Theta: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K . Replace G by $\Theta(G)$, replace K by $\Theta(K)$, and denote the resulting inclusion into the corona again by Θ . The new K is a nontrivial normal property-(T) subgroup, since property (T) passes to quotients. It is contained in $R_{\infty \rightarrow 2}(\Theta(G))$, because an operator norm asymptotic representation of $\Theta(G)$, precomposed with the quotient map, is one of the original group.

Let $p_K \in C_{\max}^*(K)$ be the Kazhdan projection and let p be its image in $\mathcal{Q}_{\mathbf{d}}$. Under any faithful representation of the corona, p is the projection onto the K -fixed vectors. Normality of K gives

$$\Theta(g)p\Theta(g)^* = p \quad (g \in G).$$

Indeed, in a faithful representation of the corona the two projections have ranges $\Theta(g) \text{Fix } \Theta(K)$ and $\text{Fix } \Theta(K)$; these agree because $gKg^{-1} = K$. Put $q = 1 - p$. The projection q is nonzero: if $p = 1$, then every vector is fixed by K , so every element of K has trivial corona image. Lemma 2.7 gives positive integers r_n and an operator norm asymptotic representation $W_n: G \rightarrow \mathcal{U}(r_n)$. Let $\widehat{\Theta}: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{r}})$ be its corona homomorphism. Under the coordinatewise corner identifications, $\widehat{\Theta}$ is the coordinate restriction of $g \mapsto q\Theta(g)$.

Choose a finite Kazhdan set $S \subseteq K$ and a Kazhdan constant $\kappa > 0$. Represent $\mathcal{Q}_{\mathbf{r}}$ faithfully. The Kazhdan projection of K under $\widehat{\Theta}$ is zero, so $\widehat{\Theta}|_K$ has no nonzero fixed vectors, and the Kazhdan inequality gives, for the positive element

$$b = \frac{1}{|S|} \sum_{s \in S} (\widehat{\Theta}(s) - 1)^* (\widehat{\Theta}(s) - 1)$$

the inequality

$$b \geq \frac{\kappa^2}{|S|} 1.$$

The coordinate elements

$$b_n = \frac{1}{|S|} \sum_{s \in S} (W_n(s) - I_{r_n})^* (W_n(s) - I_{r_n})$$

represent b . Taking the normalized trace on $M_{r_n}(\mathbb{C})$ yields

$$\frac{1}{|S|} \sum_{s \in S} \|W_n(s) - I_{r_n}\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$

The order inequality for b says that the negative part of $b_n - (\kappa^2/|S|)I_{r_n}$ converges to zero in operator norm. Taking normalized traces gives the displayed inequality. After passing to a subsequence, one fixed $s_0 \in S$ therefore stays a positive Hilbert–Schmidt distance from I_{r_n} . This contradicts $s_0 \in K \leq R_{\infty \rightarrow 2}(G)$, established after passing to $\Theta(G)$. Thus every corona homomorphism kills K , as required. $\square$

Proof of Theorem A. For every $u \in \text{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, Corollary 2.6 gives

$$[ucu^{-1}, \ell] \in R_{\infty \rightarrow 2}(G).$$

Normality of $R_{\infty \rightarrow 2}(G)$ therefore gives

$$\mathfrak{D}_G(L) \leq R_{\infty \rightarrow 2}(G).$$

Since $K \leq \mathfrak{D}_G(L)$, Theorem 2.8 now gives $K \leq \text{Rad}_{\text{MF}}(G)$. Taking a nontrivial K gives the non-MF conclusion. Taking $K = G = \mathfrak{D}_G(L)$ gives $\text{Rad}_{\text{MF}}(G) = G$. $\square$

By Theorem 2.2, every finite-dimensional linear representation of G kills the generators of $\mathfrak{D}_G(L)$, and its kernel is normal, so it kills $\mathfrak{D}_G(L)$. Consequently, $\mathfrak{D}_G(L) = 1$ if G has a faithful finite-dimensional linear representation. The same conclusion holds if G is residually finite, because every nonidentity element then survives in a finite permutation representation. If G is amenable, its property-(T) subgroup L is finite [BHV]; then $uLu^{-1} = L$, so ucu^{-1} centralizes L for every $u \in \text{Comp}_G(L)$, and again $\mathfrak{D}_G(L) = 1$.

The argument is specific to operator norm approximation and gives no conclusion about soficity or hyperlinearity: normalized Hilbert–Schmidt approximations do not provide the operator norm control used to construct the conjugation representation in Theorem 2.5.

3. THE BINARY LEAVITT GROUP

The relation $s_0t_0 + s_1t_1 = 1$ of the binary Leavitt algebra gives an element $\tau \in H$ with $\tau L \tau^{-1} \leq L$ for a copy L of $\text{EL}_3(R)$, and a commutator $d \neq 1$ in $\mathfrak{D}_H(L)$; since d normally generates H , $\mathfrak{D}_H(L) = H$, and Theorem A applies.

3.1. The self-compression. Throughout this section $R = L_{\mathbb{F}_2}(1, 2)$ and

$$p = s_0t_0, \quad q = s_1t_1.$$

The relations (2) give

$$(4) \quad p + q = 1, \quad t_1qs_1 = 1.$$

In particular $q \neq 0$.

We use indices $0, \dots, 11$ for 12×12 matrices. We identify $\mathrm{GL}_3(R)$ with the subgroup of $\mathrm{GL}_{12}(R)$ consisting of the matrices $\mathrm{diag}(A, I_9)$. Define a map $\Psi: M_3(R) \rightarrow M_3(R)$ by

$$(5) \quad \Psi(A) = qI_3 + s_0At_0.$$

Here s_0At_0 denotes the matrix $(s_0a_{ij}t_0)_{ij}$. The relations $q^2 = q$, $qs_0 = t_0q = 0$, and $t_0s_0 = 1$, all consequences of (2), show that Ψ is multiplicative. Moreover,

$$\Psi(I_3) = qI_3 + s_0I_3t_0 = qI_3 + pI_3 = (p+q)I_3 = I_3,$$

so Ψ preserves the identity. Finally, $t_0\Psi(A)s_0 = A$, so Ψ is injective. Hence Ψ restricts to an injective group homomorphism $\mathrm{GL}_3(R) \rightarrow \mathrm{GL}_3(R)$.

To realize Ψ by conjugation on the embedded copy of $\mathrm{GL}_3(R)$, define the following 6×6 block matrices:

$$X = \begin{pmatrix} s_0I_3 & s_1t_0I_3 \\ 0 & t_1I_3 \end{pmatrix}, \quad Y = \begin{pmatrix} t_0I_3 & 0 \\ s_0t_1I_3 & s_1I_3 \end{pmatrix}.$$

A block multiplication using (2) gives $Y = X^{-1}$. Put

$$(6) \quad \tau = \mathrm{diag}(X, Y) \in \mathrm{GL}_{12}(R).$$

A direct block multiplication gives

$$(7) \quad \tau \mathrm{diag}(A, I_9) \tau^{-1} = \mathrm{diag}(\Psi(A), I_9).$$

Lemma 3.1. *The matrix τ defined in (6) belongs to $\mathrm{EL}_{12}(R)$.*

Proof. For $B \in M_6(R)$, the block matrices $\begin{pmatrix} I & B \\ 0 & I \end{pmatrix}$ and $\begin{pmatrix} I & 0 \\ B & I \end{pmatrix}$ lie in $\mathrm{EL}_{12}(R)$: they are the products of the elementary matrices $e_{r,6+s}(b_{rs})$, respectively $e_{6+r,s}(b_{rs})$, and no cross terms arise because distinct off-diagonal matrix units have product zero. Whitehead's identity

$$\mathrm{diag}(X, X^{-1}) = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -X^{-1} & I \end{pmatrix} \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}$$

with $I = I_6$ therefore expresses τ as a product of six matrices of $\mathrm{EL}_{12}(R)$. $\square$

Set $H = \mathrm{EL}_{12}(R)$, and let $L = \mathrm{EL}_3(R) \leq H$ denote the image of the embedding $A \mapsto \mathrm{diag}(A, I_9)$.

Proposition 3.2. *The groups H and L have property (T), and $\tau \in H$ satisfies*

$$(8) \quad \tau L \tau^{-1} \leq L.$$

Proof. The ring R is generated by s_0, s_1, t_0, t_1 as a unital ring. The theorem of Ershov and Jaikin-Zapirain therefore gives property (T) for $\mathrm{EL}_{12}(R)$ and $\mathrm{EL}_3(R)$ [EJZ, Theorem 1.1]. Lemma 3.1 gives $\tau \in H$.

For $i \neq j$ and $a \in R$, equations (7) and (5) give

$$\tau e_{ij}(a) \tau^{-1} = e_{ij}(s_0at_0) \in L,$$

where the elementary matrices are viewed in the embedded copy of L . These matrices generate L , so (8) follows. $\square$

3.2. Simplicity and the full MF radical.

Proposition 3.3. *The group $H = \text{EL}_{12}(R)$ is nontrivial and simple.*

Proof. The ring R is purely infinite simple [AA], and therefore is an exchange ring [Ara]. Let $N \trianglelefteq H$. Since $H = \text{EL}_{12}(R)$, the subgroup $N \leq \text{GL}_{12}(R)$ is normalized by $\text{EL}_{12}(R)$. Preusser's sandwich theorem [Pre, Theorem 3] provides an ideal $I \trianglelefteq R$ such that

$$\text{EL}_{12}(R, I) \leq N \leq C_{12}(R, I).$$

Simplicity of R gives $I = 0$ or $I = R$. If $I = R$, then $\text{EL}_{12}(R, I) = H$, so $N = H$.

Suppose that $I = 0$. Then $N \leq C_{12}(R, 0) = Z(\text{GL}_{12}(R))$. A matrix that commutes with every $e_{ij}(1)$ is a scalar matrix λI_{12} . Commutation with $e_{ij}(a)$ for arbitrary $a \in R$ then gives $\lambda a = a\lambda$, so $\lambda \in Z(R)^\times$. Since $Z(R) = \mathbb{F}_2$ [AC, Corollary 4.3], we have $\lambda = 1$ and $N = 1$. Finally, $e_{01}(1) \neq 1$, so H is nontrivial. $\square$

Proposition 3.4. *Let*

$$c = e_{34}(1), \quad \ell = e_{12}(1), \quad d = [\tau c \tau^{-1}, \ell]$$

in H . Then

$$c \in C_H(L), \quad d = e_{02}(q) \neq 1, \quad \langle\langle d \rangle\rangle_H = H.$$

Proof. For $i, j \in \{0, 1, 2\}$ with $i \neq j$ and $a \in R$, we have $j \neq 3$ and $i \neq 4$. Hence the Steinberg relations give

$$[e_{ij}(a), e_{34}(1)] = 1.$$

These matrices generate L , so $c = e_{34}(1)$ belongs to $C_H(L)$.

Direct multiplication of the matrices in (6) gives

$$(9) \quad \tau c \tau^{-1} = e_{01}(q) e_{34}(1).$$

The second factor commutes with $\ell = e_{12}(1)$. The identity $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ therefore gives $d = e_{02}(q)$, which is nontrivial because $q \neq 0$.

Let $N = \langle\langle d \rangle\rangle_H$. Since $t_1 q s_1 = 1$, the Steinberg relations give

$$[d, e_{23}(s_1)] = e_{03}(q s_1) \in N, \quad [e_{10}(t_1), e_{03}(q s_1)] = e_{13}(1) \in N.$$

Conjugation by elementary Weyl matrices gives $e_{ij}(1) \in N$ for every $i \neq j$. For $a \in R$ and distinct i, j, k ,

$$[e_{ik}(a), e_{kj}(1)] = e_{ij}(a) \in N.$$

Thus N contains every elementary generator of H , so $N = H$. $\square$

Proof of Theorem B. Let Y be the set of elements $e_{ij}(a)$ with $i \neq j$ and $a \in \{1, s_0, s_1, t_0, t_1\}$. Since $e_{ij}(a + b) = e_{ij}(a)e_{ij}(b)$ and $[e_{ik}(a), e_{kj}(b)] = e_{ij}(ab)$ for distinct i, j, k , and since every element of R is a sum of monomials in s_0, s_1, t_0, t_1 , induction on the length of a monomial shows that every $e_{ij}(a)$ lies in $\langle Y \rangle$; there is always a third index because $12 \geq 3$. Thus $H = \langle Y \rangle$ is finitely

generated, and H has property (T) by the Ershov–Jaikin–Zapirain theorem for elementary groups over finitely generated rings [EJZ, Theorem 1.1].

The containment (8), the centrality of c relative to L , and Proposition 3.4 show that $d \in \mathfrak{D}_H(L)$. The subgroup $\mathfrak{D}_H(L)$ is normal and Proposition 3.4 gives $\langle\langle d \rangle\rangle_H = H$, so $\mathfrak{D}_H(L) = H$. The group H has property (T), so Theorem A, applied with $K = H$, gives $\text{Rad}_{\text{MF}}(H) = H$. Proposition 3.3 shows that H is nontrivial and simple.

If M is MF and $f: H \rightarrow M$ is a homomorphism, compose f with a faithful corona embedding of M . Since $\text{Rad}_{\text{MF}}(H) = H$, this composition and hence f are trivial. If H were MF, its identity homomorphism would contradict this conclusion.

Finally, $C_r^*(H)$ is separable because H is countable. Its canonical trace is faithful, so it is stably finite. If an MF embedding $e: C_r^*(H) \rightarrow \mathcal{Q}_{\mathbf{d}}$ existed and $p = e(1)$, then $h \mapsto e(\lambda_h) + (1 - p)$ would embed H in $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$. This contradicts the fact that H is not MF. $\square$

4. A TORSION-FREE GROUP WITH FULL MF RADICAL

A group is acylindrically hyperbolic if it admits a non-elementary acylindrical action on a hyperbolic space [Os]. For such a group G , Hull [Hu, Theorem 3.12] provides a generating set A such that the Cayley graph $\Gamma(G, A)$ is hyperbolic and the action of G on it is acylindrical and non-elementary; a subgroup $N \leq G$ is *suitable* with respect to A if N acts non-elementarily on $\Gamma(G, A)$ and normalizes no nontrivial finite subgroup of G [Hu, Definition 1.4]. We use Hull’s small cancellation theorem [Hu, Theorem 7.1] in the following form.

Theorem 4.1 (Hull’s small cancellation theorem). *Let G be acylindrically hyperbolic, let $N \leq G$ be suitable with respect to A , and let $t_1, \dots, t_m \in G$ and a finite subset $F \subseteq G$ be given. Then there is a surjective homomorphism $q: G \rightarrow Q$ such that Q is acylindrically hyperbolic, $q|_F$ is injective, $q(t_i) \in q(N)$ for all i , and every element of finite order in Q is the image of an element of the same order in G .*

Moreover the kernel of q is the normal closure of m elements; this is read off from Hull’s proof: the case $m = 1$ adds the single relator $W = t_1^{-1}h_1^{m_1}h_2^{l_1} \cdots h_1^{m_n}h_2^{l_n}$ for suitable $h_1, h_2 \in N$, and the general case is an induction on m .

Lemma 4.2 (saturation). *Let G be finitely presented, torsion-free, and acylindrically hyperbolic, let $N \trianglelefteq G$ be nontrivial, and let $F \subseteq G$ be finite. Then there is a surjective homomorphism $q: G \rightarrow Q$ such that Q is two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $q|_F$ is injective, and $q(N) = Q$.*

Proof. Fix A as above. Since G is torsion-free and $N \neq 1$, N is infinite, so $|N \cap gNg^{-1}| = \infty$ for every $g \in G$, and by Osin [Os, Lemma 7.1] N acts non-elementarily on $\Gamma(G, A)$. The only finite subgroup of G is trivial, so

N is suitable. By Hull [Hu, Corollary 5.7 and Lemma 5.8], N contains two elements h_1, h_2 such that $N_0 = \langle h_1, h_2 \rangle$ is again suitable. Apply Theorem 4.1 to N_0 with $t_1, \dots, t_m$ a finite generating set of G and with the set F . Then

$$Q = \langle q(t_1), \dots, q(t_m) \rangle \leq q(N_0) \leq q(N) \leq Q,$$

so $Q = q(N_0) = \langle q(h_1), q(h_2) \rangle$ and $q(N) = Q$. The group Q is torsion-free because G is, and finitely presented because it is G with m relators added. $\square$

Following Fournier-Facio [FF, §2], let U be a universal finitely presented torsion-free group, one containing a copy of every finitely presented torsion-free group [Chi]; let S be a finitely presented infinite simple torsion-free group [HL]; and let H_0 be a torsion-free hyperbolic group with property (T), obtained from the density model at a parameter between $1/3$ and $1/2$ [KK, Ol]. Small cancellation over the relatively hyperbolic pair $(U * H_0, U)$ embeds U in a finitely presented torsion-free quotient P of H_0 [FF2, Proposition 2.3]; torsion-freeness is preserved by Osin's embedding theorem [OsSC, Theorem 2.4(5)]. Consequently, P has property (T). By universality, P contains a subgroup $P_1 \times P_2 \times S$ with $P_i \cong P$. Let E be the double HNN extension of P with stable letters u_1, u_2 and $u_i P u_i^{-1} = P_i$; it is finitely presented and torsion-free, and its Bass-Serre action makes it acylindrically hyperbolic by the tree criterion of Minasyan-Osin [MO]. Hull's common quotient theorem [Hu, Corollary 7.4] applied to E and H_0 gives a surjection $\pi: E \rightarrow G_0$ onto a finitely presented torsion-free group with property (T), injective on a prescribed finite subset of E ; since E is finitely generated, the proof of that corollary produces G_0 acylindrically hyperbolic. We run the construction with the prescribed finite subset containing an element $s \neq 1$ of S . Then $\pi(S) \neq 1$, so $\pi|_S$ is injective because S is simple.

Put $\Gamma = \pi(P)$, $t = \pi(u_1)$, and $J = t^{-1}\pi(S)t$. Since S commutes with $P_1 = u_1 P u_1^{-1}$, the subgroup J centralizes Γ , and $t\Gamma t^{-1} = \pi(P_1) \leq \Gamma$, so $t \in \text{Comp}_{G_0}(\Gamma)$. Moreover $tJt^{-1} = \pi(S) \leq \Gamma$, and Γ has property (T) as a quotient of P .

Lemma 4.3. *Let $\rho: G_0 \rightarrow L$ be a surjective homomorphism with $\rho(\pi(S)) \neq 1$. Then $\rho(\pi(S)) \leq \mathfrak{D}_L(\rho(\Gamma))$.*

Proof. Write $\Gamma_L = \rho(\Gamma)$, $t_L = \rho(t)$, and $J_L = \rho(J)$. Then $t_L \Gamma_L t_L^{-1} \leq \Gamma_L$, J_L centralizes Γ_L , and $t_L J_L t_L^{-1} = \rho(\pi(S)) \leq \Gamma_L$. For $c \in J_L$ and $\ell \in \Gamma_L$, the commutator $[t_L c t_L^{-1}, \ell]$ lies in $\mathfrak{D}_L(\Gamma_L)$ by (1). As c ranges over J_L , the element $t_L c t_L^{-1}$ ranges over $\rho(\pi(S))$, and ℓ may be taken in $\rho(\pi(S)) \leq \Gamma_L$; hence the commutator subgroup of $\rho(\pi(S))$ lies in $\mathfrak{D}_L(\Gamma_L)$. Since $\rho(\pi(S)) \neq 1$ and S is simple, $\rho(\pi(S)) \cong S$ is perfect, so $\rho(\pi(S)) \leq \mathfrak{D}_L(\Gamma_L)$. $\square$

Proof of Theorem C. Let N be the normal closure of $\pi(S)$ in G_0 . It is nontrivial, so Lemma 4.2 applied to G_0 , N , and $F = \{1, \pi(s)\}$ gives $q: G_0 \rightarrow Q$ with Q two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $q(N) = Q$, and $q(\pi(s)) \neq 1$. The group Q has property (T) as a quotient of G_0 .

Let $r: Q \rightarrow L$ be a surjection onto a nontrivial group and put $\rho = r \circ q$. Since $q(N) = Q$, the normal closure of $\rho(\pi(S))$ in L is $\rho(N) = L$; as $L \neq 1$, this forces $\rho(\pi(S)) \neq 1$. By Lemma 4.3, $\rho(\pi(S)) \leq \mathfrak{D}_L(\rho(\Gamma))$, and since $\mathfrak{D}_L(\rho(\Gamma))$ is normal in L and contains a normal generating set of L , it equals L . The group L has property (T) as a quotient of Q , and $\rho(\Gamma)$ has property (T) as a quotient of Γ . Theorem A, applied to L with the subgroup $\rho(\Gamma)$ and $K = L$, gives $\text{Rad}_{\text{MF}}(L) = L$. Taking r the identity gives $\text{Rad}_{\text{MF}}(Q) = Q$, and a nontrivial group equal to its own MF radical is not MF. $\square$

ACKNOWLEDGMENTS

The author thanks Francesco Fournier-Facio for early advice and detailed comments on earlier drafts, and Alon Dogon for organizing the ensuing discussion and for comments on the surrounding literature. The author also thanks Tatiana Shulman for clarifying the terminology and historical context of the MF question, and Caleb Eckhardt for discussions of MF representations and matrix norms.

REFERENCES

- [AA] G. Abrams and G. Aranda Pino, *Purely infinite simple Leavitt path algebras*, J. Pure Appl. Algebra **207** (2006), no. 3, 553–563, [doi:10.1016/j.jpaa.2005.10.010](#).
- [AW] C. A. Akemann and M. E. Walter, *Unbounded negative definite functions*, Canad. J. Math. **33** (1981), no. 4, 862–871.
- [AT] V. Alekseev and A. Thom, *Centralizers of sofic approximations of Kazhdan groups*, preprint, 2026, [arXiv:2608.05362](#).
- [AC] G. Aranda Pino and K. Crow, *The center of a Leavitt path algebra*, Rev. Mat. Iberoam. **27** (2011), no. 2, 621–644, [doi:10.4171/RMI/649](#).
- [Ara] P. Ara, *The exchange property for purely infinite simple rings*, Proc. Amer. Math. Soc. **132** (2004), no. 9, 2543–2547, [doi:10.1090/S0002-9939-04-07369-1](#).
- [BHV] B. Bekka, P. de la Harpe, and A. Valette, *Kazhdan’s Property (T)*, Cambridge University Press, 2008.
- [BDL] B. Bachner, A. Dogon, and A. Lubotzky, *On L^1 -approximation of groups*, J. Algebra **702** (2026), 235–243, [doi:10.1016/j.jalgebra.2026.04.041](#), [arXiv:2508.17392](#).
- [BK] B. Blackadar and E. Kirchberg, *Generalized inductive limits of finite-dimensional C^* -algebras*, Math. Ann. **307** (1997), no. 3, 343–380, [doi:10.1007/s002080050039](#).
- [CDE] J. R. Carrión, M. Dadarlat, and C. Eckhardt, *On groups with quasidiagonal C^* -algebras*, J. Funct. Anal. **265** (2013), no. 1, 135–152, [doi:10.1016/j.jfa.2013.04.004](#).
- [Chi] M. Chiodo, *On torsion in finitely presented groups*, Groups Complex. Cryptol. **6** (2014), no. 1, 1–8, [arXiv:1107.1489](#).
- [Dad] M. Dadarlat, *Obstructions to matricial stability of discrete groups and almost flat K -theory*, Adv. Math. **384** (2021), Paper No. 107722, [doi:10.1016/j.aim.2021.107722](#).
- [DGLT] M. De Chiffre, L. Glebsky, A. Lubotzky, and A. Thom, *Stability, cohomology vanishing, and nonapproximable groups*, Forum Math. Sigma **8** (2020), Paper No. e18, [doi:10.1017/fms.2020.5](#).
- [EJZ] M. Ershov and A. Jaikin-Zapirain, *Property (T) for noncommutative universal lattices*, Invent. Math. **179** (2010), 303–347, [doi:10.1007/s00222-009-0218-2](#).
- [FF] F. Fournier-Facio, *A torsion-free non-sofic group*, preprint, 2026, [arXiv:2608.02025](#).
- [FF2] F. Fournier-Facio, *Stability, approximable quotients, and higher property (T)*, preprint, 2025, [arXiv:2512.09180](#).

- [GKM] D. Gao, S. Kunnawalkam Elayavalli, A. Manzoor, and G. Patchell, *A new source of purely finite matricial fields*, preprint, 2026, [arXiv:2603.24502](#).
- [GH] I. Goldbring and B. Hart, *The universal theory of the hyperfinite II_1 factor is not computable*, Bull. Symbolic Logic **30** (2024), no. 2, 181–198, [doi:10.1017/bsl.2024.7](#), [arXiv:2006.05629](#).
- [Hu] M. Hull, *Small cancellation in acylindrically hyperbolic groups*, Groups Geom. Dyn. **10** (2016), no. 4, 1077–1119, [doi:10.4171/GGD/377](#).
- [HL] J. Hyde and Y. Lodha, *Finitely presented simple left-orderable groups in the landscape of Richard Thompson’s groups*, Ann. Sci. Éc. Norm. Supér. (4) **58** (2025), no. 2, 419–432, [arXiv:2302.04805](#).
- [JNVWY] Z. Ji, A. Natarajan, T. Vidick, J. Wright, and H. Yuen, $\text{MIP}^* = \text{RE}$, Commun. ACM **64** (2021), no. 11, 131–138, [doi:10.1145/3485628](#), [arXiv:2001.04383](#).
- [K–T] H. V. Khanh and V. H. Thanh, *Matrix generators for the unit groups of $L_K(1, d)$* , preprint, 2026, [arXiv:2607.10351](#).
- [Kor] A. I. Korchagin, *The MF-property for countable discrete groups*, Math. Notes **102** (2017), no. 2, 198–211, [doi:10.1134/S0001434617070227](#), [arXiv:1704.06906](#).
- [KK] M. Kotowski and M. Kotowski, *Random groups and property (T): Żuk’s theorem revisited*, J. Lond. Math. Soc. (2) **88** (2013), no. 2, 396–416, [arXiv:1106.2242](#).
- [Lev] W. G. Leavitt, *The module type of a ring*, Trans. Amer. Math. Soc. **103** (1962), 113–130, [doi:10.1090/S0002-9947-1962-0132764-X](#).
- [LO] A. Lubotzky and I. Oppenheim, *Non p -norm approximated groups*, J. Anal. Math. **141** (2020), no. 1, 305–321, [doi:10.1007/s11854-020-0119-2](#).
- [MO] A. Minasyan and D. Osin, *Acylindrical hyperbolicity of groups acting on trees*, Math. Ann. **362** (2015), no. 3–4, 1055–1105, [arXiv:1310.6289](#).
- [KT] G. Kun and A. Thom, *Inapproximability of actions and Kazhdan’s property (T)*, preprint, 2019, [arXiv:1901.03963](#).
- [OAI] OpenAI, *Nonsofic groups exist*, in *Ten advances in mathematics and theoretical computer science*, Chapter 3, 78–95, 2026, <https://cdn.openai.com/pdf/ten-proofs-oai.pdf>.
- [OI] Y. Ollivier, *Sharp phase transition theorems for hyperbolicity of random groups*, Geom. Funct. Anal. **14** (2004), no. 3, 595–679, [arXiv:math/0301187](#).
- [Os] D. Osin, *Acylindrically hyperbolic groups*, Trans. Amer. Math. Soc. **368** (2016), no. 2, 851–888, [doi:10.1090/tran/6343](#).
- [OsSC] D. Osin, *Small cancellations over relatively hyperbolic groups and embedding theorems*, Ann. of Math. (2) **172** (2010), no. 1, 1–39, [arXiv:math/0411039](#).
- [Pre] R. Preusser, *On general linear groups over exchange rings*, Linear and Multilinear Algebra **70** (2022), no. 4, 705–713, [doi:10.1080/03081087.2020.1743636](#).
- [Sch] C. Schafhauser, *Finite dimensional approximations of certain amalgamated free products of groups*, preprint, 2023, [arXiv:2306.02498](#).
- [Sh] T. Shulman, *The MF property for amalgamated free products*, preprint, 2026, [arXiv:2603.13564](#).
- [Th] A. Thom, *Finitary approximations of groups and their applications*, in *Proceedings of the International Congress of Mathematicians—Rio de Janeiro 2018*, Vol. III: Invited Lectures, World Scientific, Hackensack, NJ, 2018, 1779–1799, [doi:10.1142/9789813272880_0117](#), [arXiv:1712.01052](#).